



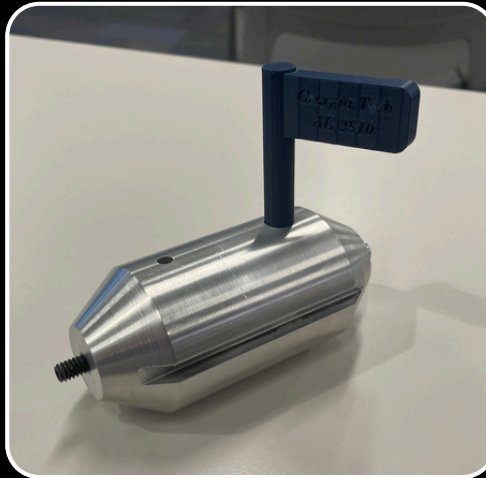
AE 2810 Presentation Card

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Final Part

Space Module
AE2810 Flag



Matching

- Overall, my part matches the initial drawing closely.
- All of the critical measurements are within the specified tolerance.
- One minor issue is that one of the holes is not perfectly circular; it contains some external formation/scratches.

Lessons

- 1) Meeting and satisfying tolerances is key for macro-manufacturing success.
- 2) Reviewing drawings beforehand and having a plan is needed for smooth execution.
- 3) Don't overthink, but instead tackle the process.

Reflection

Transforming from a novice to becoming proficient in operating machinery has been a profound experience.

I was most surprised by the precision required to translate a digital concept into a tangible part.

Practicing the “design for manufacturability” rule is a lesson that will undoubtedly refine my approach to engineering design as I continue on with my career.

I leave this course with not only new technical skills, but a much keener eye for detail.

PLANNED

Manufacturing Process

ACTUAL

1. Face off, flip, cut to length (M)
2. Use drill bit to cut slit after locating edges (M)
3. Drill 2 through holes and tap 1 of them (M)
4. Flip and counterbore the other hole (M)
5. Drill hole in side and tap (L)
6. Compound angle and drill both sides (L)
7. Turn off knob (L)
8. Sandblast and polish

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